

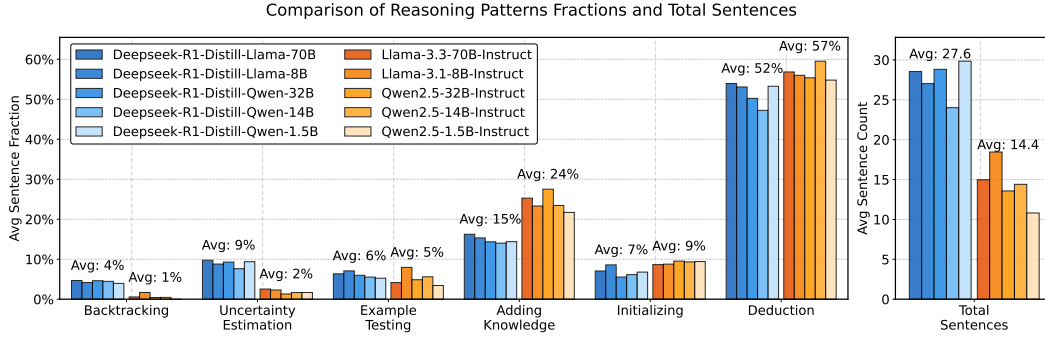


Figure 2: Comparison of behavioral patterns across five DeepSeek-R1-Distill models and five baseline models on 100 randomly selected tasks from our dataset (cf. Section 4.1). The plot on the left shows the fraction of sentences annotated with each behavioral category. The plot on the right shows the average number of sentences per response. Thinking models generate substantially longer responses (27.6 vs 14.4 sentences on average) and exhibit a higher fractions of backtracking, uncertainty estimation and example testing behaviors, but lower fractions of knowledge augmentation.

- **Uncertainty Estimation:** The model explicitly states its confidence or uncertainty regarding its reasoning.
- **Backtracking:** The model abandons its current approach and explores an alternative strategy.

An annotated example of these behaviors is provided in Appendix B. To systematically annotate reasoning chain in our experiments, we employ GPT-4o (see Appendix A for details). To quantify the prevalence of these reasoning behaviors, we generate reasoning chains using five DeepSeek-R1-Distill models (Qwen-14B, Qwen-32B, Llama-70B, Qwen-1.5B, and Llama-8B) and compare them against five baseline models (Qwen2.5-Math-1.5B, Qwen2.5-32B-Instruct, Qwen2.5-14B-Instruct, Llama-3.3-70B-Instruct, and Llama-3.1-8B-Instruct) across 100 diverse tasks (see Section 4.1). We then automatically annotate each reasoning chain and compute the fraction of sentences annotated with each behavioral category. Results are presented in Figure 2.

The results reveal several key distinctions between thinking and baseline models. Most notably, thinking models generate substantially longer responses, averaging 27.6 sentences vs 14.4 for baseline models. Both model types devote similar fractions of their responses to initialization and deduction, suggesting that these processes are comparable across model types. In contrast, baseline models show lower fractions of backtracking, uncertainty estimation, and, to a lesser extent, example testing, instead allocating more of their responses to knowledge augmentation.

These patterns suggest that uncertainty estimation and backtracking are prominent behaviors that distinguish thinking models, followed by example testing and knowledge addition. Given that initialization and deduction do not seem to be behaviors specific to thinking models, steering on these seems conceptually ill-defined; initialization occurs once at the start and deduction is fundamental to any LLM, therefore we omit them from the empirical study on steering vectors.

4 EXTRACTING AND EVALUATING STEERING VECTORS

In this section, we demonstrate that the reasoning behaviors of the DeepSeek-R1-Distill models characterized in Section 3 are mediated by linear directions, the *steering vectors*. We assess the causal effect of these vectors by comparing the model’s original reasoning chains to those generated under positive and negative steering (adding and subtracting the steering vectors). Our findings indicate that the DeepSeek-R1-Distill models have distinct mechanisms to achieve their reasoning process. Additionally, our steering vectors provide an efficient way to influence these models’ reasoning behavior, for example, increasing their tendency to backtrack or modulate their inherent uncertainty in their own reasoning.

Category	Description
Mathematical Logic	Problems requiring formal logical operations, mathematical proofs, and numerical reasoning
Spatial Reasoning	Tasks involving visualization, geometric manipulation, and understanding spatial relationships
Verbal Logic	Problems focused on language-based reasoning, syllogisms, and verbal analogies
Pattern Recognition	Questions requiring identification and continuation of sequences or abstract patterns
Lateral Thinking	Problems that require creative, non-linear approaches to reach unconventional solutions
Causal Reasoning	Tasks involving understanding cause-and-effect relationships and making causal inferences
Probabilistic Thinking	Problems requiring reasoning about uncertainty, probability, and statistical concepts
Systems Thinking	Questions about complex systems, interconnected components, and emergent behaviors
Creative Problem Solving	Open-ended problems requiring novel approaches and innovative solutions
Scientific Reasoning	Tasks involving hypothesis formation, experimental design, and evidence evaluation

Table 1: Task categories used to analyze reasoning behaviors.

4.1 EXPERIMENTAL SETUP

For our experiments, we generate a dataset of 500 tasks across 10 categories using Claude 3.5 Sonnet (see Table 1). We conduct our experiments on three DeepSeek-R1-Distill models: Qwen-14B, Qwen-1.5B, and Llama-8B. When generating a reasoning chain, we use greedy decoding and 1000 max tokens per response.

4.2 LOCATING CAUSALLY RELEVANT ACTIVATIONS

To extract robust steering vectors, we first identify the activations where these vectors are linearly represented within the model. We focus on the *residual stream activations*, i.e., the outputs of each transformer layer. Given a reasoning chain generated by a DeepSeek-R1-Distill model, we identify both the *token positions* and *layers* where the steering vector is active. This process consists of two key steps:

1. **Identifying relevant token positions:** Determine which tokens in the reasoning chain correspond to a specific behavioral category.
2. **Determining causally relevant layers:** Use *attribution patching* (Section 2) to evaluate which layers contribute causally to the behavior in question.

To obtain token positions associated with each behavioral category, we generate 500 reasoning chains with the tasks introduced in Section 4.1, using both DeepSeek-R1-Distill models and then annotate them automatically with GPT-4o. Since the DeepSeek-R1-Distill models are autoregressive, we consider for each category both the token position **preceding** the start of a token-sequence annotated with the current category and the annotated sequence itself as the causally relevant token positions (up to 10 tokens). This ensures that we capture both the decision point where the model transitions into the behavior and the behavior’s execution phase.

To identify the causally relevant layers for each behavioral category, we first extract a steering vector candidate from every layer using the *Difference of Means* method (Section 2.2):

$$\mathbf{u}_\ell^c = \frac{1}{|D_+|} \sum_{p_i \in D_+} \bar{\mathbf{a}}_\ell^c(p_i) - \frac{1}{|D_-|} \sum_{p_j \in D_-} \mathbf{a}_\ell^c(p_j), \quad \text{with} \quad \bar{\mathbf{a}}_\ell^c(p_i) = \frac{1}{|\text{seq}_c(p_i)|} \sum_{t \in \text{seq}_c(p_i)} \mathbf{a}_\ell(t).$$

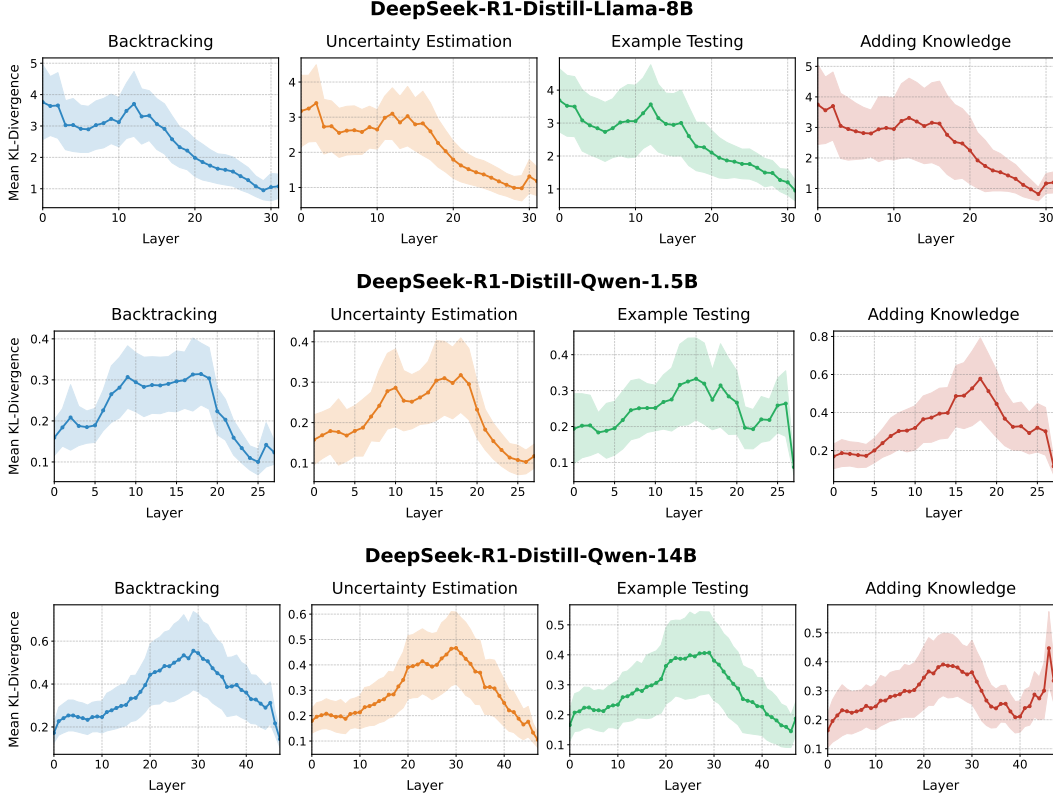


Figure 3: Causal impact of candidate steering vectors across model layers. The y-axis represents the absolute mean KL-divergence for the next-token logit distribution when removing the steering vector at each layer. The steering vectors for all reasoning mechanisms have similar peaks in the middle layers of the respective models.

where:

- $\mathbf{a}_\ell(t)$ represents the residual stream activation at layer ℓ for token position t .
- $\text{seq}_c(p)$ is the set of all token sequences within prompt p that are annotated with category c , including the preceding token position.
- $\bar{\mathbf{a}}_\ell^c(p_i)$ denotes the *mean activation* across all token positions within the annotated sequences of category c at layer ℓ .
- D_+ consists of prompts containing at least one sequence labeled with category c , while D_- represents the full dataset.

The resulting vector $\mathbf{u}_\ell^c$ serves as a candidate steering vector for each layer. To ensure consistent scaling across different sequence lengths and behaviors, we normalize each steering vector to have the same magnitude as the mean overall activation:

$$\mathbf{u}_\ell^{c,\text{norm}} = \mathbf{u}_\ell^c \cdot \frac{\|\bar{\mathbf{a}}_\ell^{\text{overall}}\|}{\|\mathbf{u}_\ell^c\|}$$

where $\bar{\mathbf{a}}_\ell^{\text{overall}}$ is the mean activation across all tokens in the dataset at layer ℓ . This normalization helps account for varying sequence lengths and ensures steering vectors have comparable magnitudes across different behavioral categories.